Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Goldberg, Alfred L Ph.D.		
Project Title:	The Activation of Protein Breakdown in Muscle Upon Unloading and Possible Countermeasures		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5817	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/28/2007
No. of Post Docs:	10	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-MA00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The rapid loss of muscle mass that occurs in astronauts in space due to muscle unloading and in patients with many systemic diseases results primarily from accelerated degradation of muscle proteins. This enhancement of protein breakdown is mainly due to activation of the Ub-proteasome pathway. Our major goal is to clarify the biochemical basis for the activation of this proteolytic pathway and thus to develop pharmacological agents that reduce this excessive degradation and retard muscle wasting. Recently, we identified a set of genes ("atrogins") whose expression increases or decreases coordinately when muscles atrophy. In order to achieve a fuller understanding of the atrophy process, we plan to further study this transcriptional program and its regulation, especially after disuse. Of particular importance was the recent finding that the two genes induced most dramatically in atrophying muscles are the muscle-specific ubiquitin ligases (E3s), atrogin-1 (MAFbx) and MuRF1. If either of these enzymes is knocked out, the extent of muscle wasting is reduced. These Ub-ligases thus are very attractive therapeutic targets. To fully understand the initiation of the atrophy process, and to develop rational countermeasures, we are studying the signal transduction systems that activate transcription of these genes in simple models of muscle atrophy in cultured myotubes. We recently found that the key factor in muscle hypertrophy, IGF-1, rapidly suppresses the expression of atrogin-1 and MuRF1 and prevents their induction by glucocorticoids. These effects of IGF-1 appear to be mediated by the PI3-kinase-AKT pathway, which probably inactivates one of the Forkhead transcription factors. Our primary goals will be to identify the precise steps in this kinase cascade and the key transcription through which disuse and glucocorticoids activate and IGF-1 inhibits expression of atrogin-1 and MuRF1. We shall also examine whether other atrogins are regulated by the same signaling systems and transcription factor. These studies should identify novel therapeutic targets (e.g. key kinases) whose inhibition blocks atrogin-1 and MuRF1 induction. Using these enzymes and "reporter gene" constructs, we shall screen libraries of small molecules for agents that prevent induction of atrogin-1 and MuRF1. Inhibiting the expression of these key ligases represents an exciting new therapeutic approach to prevent muscle wasting in space personnel and in diverse disease states.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If successful, this research may yield novel therapies to prevent or treat the marked wasting of muscle seen in various bed-ridden patients and ones with systemic disease (cardiac failure, cancer, sepsis, AIDS).
Task Progress:	See "Main Findings"
Bibliography Type:	Description: (Last Updated: 09/01/2017)
Articles in Peer-reviewed Journals	Lecker, S. H., R. T. Jagoe, A. Gilbert, M. Gomes, V. Baracos, J. Bailey, S. R. Price, W. E. Mitch, and A. L. Goldberg "Multiple types of skeletal muscle atrophy involve a common program of changes in gene expression. FASEB J" 39-51 , Jan-2004
Articles in Peer-reviewed Journals	Sacheck, J. M., A. Ohtsuka, S. C. Mc Lary, and A. L. Goldberg "IGF-1 stimulates muscle growth by suppressing protein breakdown and expression of atrophy-related ubiquitin ligases, atrogin-1 and MuRF1 Am J Physiol Endocrynol" E591-601 , Apr-2004
Articles in Peer-reviewed Journals	Sandri, M., C. Sandri, A. Gilbert, C. Skurk, E. Calabria, A. Picard, K. Walsh, S. Schiaffino, S. H. Lecker, and A. L. Goldberg "Foxo transcription factors induce the atrophy-related ubiquitin ligase atrogin-1 and cause skeletal muscle atrophy Cell" 399-412 , Apr-2004
Awards	" 2004 Plenary Lecture of the American Society of Nephrology " Jan-2005
Awards	" Annual Fred Fay Lecture (Univ Mass School of Medicine) " Jan-2005
Awards	" Dr. Marco Sandri was selected for one of two Telethon Professorships in Italy. He will use it at the University of Padua School of Medicine. " Jan-2005
Awards	" One of the 2004 Nobel Lectures (Karolinska Institute, Stockholm, Sweden) " Jan-2005
Awards	" Severo Ochoa Lecture (New York University School of Medicine) " Jan-2005
Books/Book Chapters	Jagoe, R. T., N. E. Tawa, Jr., and A. L. Goldberg "Protein and amino acid metabolism in muscle Myology" Ed. Engel, A., and C. Franzini-Armstrong McGraw-Hill, 535-564 , Jan-2005
Patents	Goldberg, A. L., M. Sandri and S. H. Lecker "Foxo transcription factors are novel pharmacological targets to inhibit skeletal muscle atrophy. " Jan-2005
Presentation	Goldberg, A. L. "Functions of the Proteasome in Protein Degradation and Immune Surveillance: From Basic Understanding to Human Therapy " Oct-2004
Presentation	Goldberg, A. L. "Induction of Atrophy-Related Genes and the Ubiquitin Ligase, Atrogin-1, in Muscle during Disease States " May-2004
Presentation	Goldberg, A. L. "Mechanisms for Induction of the Key Ubiquitin Ligase, Atrogin-1, and Muscle Atrophy in Disease States " Jun-2004
Presentation	Goldberg, A. L. "Regulation of the Expression of Atrophy-Specific Ubiquitin-Ligases, Atrogin-1 and MuRF1, in Normal and Atrophying Muscle " Jun-2004
Presentation	Jagoe, R. T., J. Sacheck, S. H. Lecker, J. K. Hyatt, A. Gilbert, M. Gomes, V. Baracos, J. Bailey, S. R. Price, W. E. Mitch, R. Roy, V. R. Edgerton, and A. L. Goldberg "Multiple types of skeletal muscle atrophy involve a common program of changes in gene expression " Dec-2004
Presentation	Sandri, M., C. Sandri, A. Gilbert, S. Schiaffino, S. Lecker, and A. L. Goldberg "Foxo Transcription Factors Induce the Atrophy-Related Ubiquitin Ligase, Atrogin-1, and Cause Skeletal Muscle Atrophy " Jan-2004